

Part A – Setup and Dataset

Q1. Import the necessary libraries.

```
import pandas as pd
import numpy as np
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression, Ridge, Lasso
from sklearn.metrics import mean_squared_error
```

Q2. Create a sample dataset for salary prediction.

```
data = {
    "Experience": [1,2,3,4,5,6,7,8,9,10],
    "Age": [22,25,27,30,32,35,38,40,43,45],
    "Education_Level": [1,2,2,3,3,4,4,4,5,5],
    "Salary": [30000,35000,40000,45000,50000,55000,60000,65000,70000,75000]
}

df = pd.DataFrame(data)
print(df)
```

Part B – Data Preparation

Q3. Split the dataset into training and testing data.

Part C – Model Training

Q4. Fit a simple Linear Regression model (no regularization).

Q5. Fit a Ridge Regression model (L2 Regularization).

Q6. Fit a Lasso Regression model (L1 Regularization).

Part D – Comparison

Q7. Compare the MSE (error) across all models.

Q8. Display all coefficients together.

Part E – Visualization (Optional for Fast Students)

Q9. Create a Bar chart showing which model has the lowest error (best generalization).

Q10. What problem does regularization solve?

Q11. What is the difference between L1 and L2 regularization?

Q12. What happens when the alpha value increases?

Q13. Which model is more useful for feature selection?

Q14. Why is comparing MSE important?